

**EE5311 Digital IC Design: Assignment 9**  
**Adder trees - schematic and simulation**

1. Create a schematic for the full adder implemented in Verilog. Create a corresponding symbol and a testbench that tests it for all possible input combinations.
2. Use the full adder to construct a 16-bit ripple carry adder and create a testbench to tests it for 20 random inputs.
3. Draw schematic and testbenches for the following 16-bit tree adders using Propagate/Generate modules:
  - (a) Brent-Kung adder
  - (b) Kogge-Stone adder